Ranjith Bala
Full-Stack Developer

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Career Objective

Motivated Full-Stack Developer with a year of experience in C#, .NET (WinForms), PHP (Laravel), and modern front-end technologies. Proficient in REST APIs, cloud deployment on Azure, PostgreSQL, and Docker, with additional background in IoT (ESP32) and scalable application delivery. Seeking to leverage strong technical skills to drive dynamic IT projects through robust cloud solutions.

Technical Skills

Languages: C#, .NET, PHP, JavaScript, HTML, CSS

Frameworks: Laravel, BootStrap, WinForms

Databases: MySQL, SQLite, PostgreSQL

Virtualization Platforms: Microsoft Azure, Proxmox

Tools & Others: Git, Gitub, Gitea, Postman, REST API, Composer, Docker

Web Server: Apache2

Reverse proxy: Traefik

Build tools: Visual studio build systems, .NET CLI, MS build.

Operating Systems: Windows, Linux

Frameworks & Libraries: .NET v6.2, Entity Framework & Laravel (MVC architecture, Eloquent ORM, Blade, Artisan)

Version Control: Strong experience with Git for version control, including branching, merging, and collaboration across teams.

Testing: Proficient in writing and executing unit tests using JUnit to ensure code quality and reliability.

Monitoring & logging Tools: Grafana

Protocols: MQTT

Professional Experience

C.R.I pumps Pvt Ltd – Software Developer

■ Oct-2024 – Present | Coimbatore

- Developed robust web applications using C#, .NET (including WinForms), and PHP (Laravel), delivering responsive interfaces and scalable back-end services.
- Implemented REST APIs for seamless data integration and interoperability across distributed services.
- Deployed applications to Azure cloud, optimizing performance and uptime while leveraging containerization with Docker for streamlined delivery and scaling.
- Managed PostgreSQL databases to ensure secure, efficient data storage and retrieval, supporting

analytics and reporting needs.
- Integrated IoT solutions using ESP32, enabling real-time data acquisition and control for connected systems.

Projects

#Project 1: Optical Character Recognition (OCR) Software Solution

- Designed and developed an OCR software solution to automate text extraction from images and scanned documents.
- Utilized OCR libraries and frameworks such as Tesseract and OpenCV to achieve high accuracy in character recognition.
- Integrated preprocessing techniques like image binarization, noise reduction, and segmentation to enhance recognition performance.
- Developed a user-friendly interface to facilitate easy document upload and text extraction workflows.
- Implemented error-handling and validation mechanisms to ensure data quality and reliability.
- Enabled export of recognized text into multiple formats including plain text, PDF, and searchable documents.

#Project 2: Progressive Web Application (PWA) Development with Laravel

- Developed and deployed a Progressive Web Application using Laravel framework.
- Implemented core PWA features including service workers for offline functionality, web app manifest for home screen installation, and push notifications for enhanced user engagement.
- Leveraged Laravel's backend capabilities to create seamless, app-like user experience across mobile and desktop browsers.
- Improved application performance and reliability by enabling offline access and background content syncing.
- Ensured cross-browser compatibility and adherence to modern web standards for PWAs.

#Project 3: Production Display Software

- Developed a production monitoring software enabling users to select plant, block, shift, and line information for accurate production tracking.
- Implemented QR code scanning functionality wherein scanning captures pump serial numbers embedded in the QR codes.
- Designed logic to parse scanned data by splitting item code and serial numbers, inserting them into relevant database columns.
- Updated the scanned status column dynamically to indicate successful scanning in real-time.
- Built a dashboard to display live production counts, incrementing metrics immediately upon each successful scan and visualizing data within predefined ranges.
- Ensured seamless integration between scanning input, database updates, and dashboard visualization for efficient production monitoring.

#Project 4: Customized Sticker Printing Software

- Developed a versatile PDF-based sticker design and printing software enabling precise layout creation using X and Y coordinates.

- Designed a multi-tab interface with dedicated tabs for Text, Shapes, Images, and QR Codes to facilitate customizable sticker creation.
- In the Text tab, users input text properties including font style, size, and position coordinates for exact placement in the generated PDF.
- The Shape tab allows drawing various shapes, such as lines and rectangles, with user-defined coordinates automatically adjusted on the sticker canvas.
- Integrated an Image tab enabling users to select and embed images from their file system directly into the PDF layout.
- Facilitated dynamic generation and embedding of QR codes within sticker designs for enhanced functionality and tracking.
- Ensured real-time preview and accurate PDF generation reflecting user inputs for high-quality custom sticker printing.

Internship

Java Full Stack Development Kodnest Technologies.

- Sep-2023 – Mar-2024 | Bangalore

Education

Bachelor of Technology (Mechanical Engineering)

Annamacharya Institute of Technology & sciences, JNTUA— 2023

-Tirupati

Achievements

- Silver medal in rapid manufacturing course conducted by NPTEL (IIT Kanpur).
- Air Pollution and Control course conducted by NPTEL (IIT Roorkee).